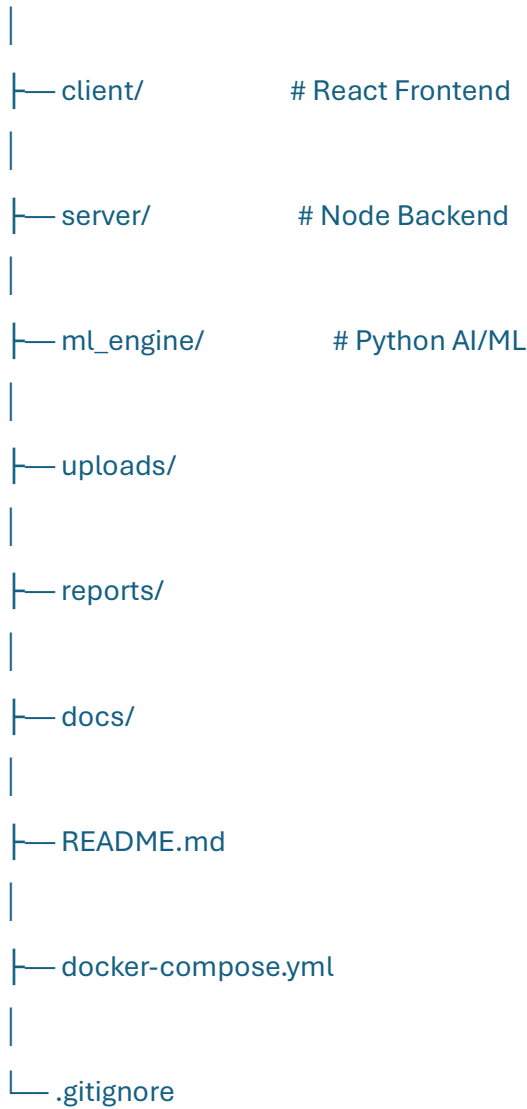


AI-Powered Data Quality & Validation Framework

Whole Project Structure

(Complete)

AI-Data-Quality-Framework/



1. Client (Frontend)

client/

```
|
|
| └─ public/
|
|
| └─ src/
|   |
|   | └─ assets/
|   |
|   | └─ components/
|   |   Navbar.jsx
|   |   Sidebar.jsx
|   |   Loader.jsx
|   |   MetricCard.jsx
|   |   DataTable.jsx
|   |   UploadBox.jsx
|   |   SeverityBadge.jsx
|   |   ChartCard.jsx
|   |
|   | └─ charts/
|   |   QualityChart.jsx
|   |   OutlierChart.jsx
|   |   MissingChart.jsx
|   |   CategoryChart.jsx
|   |
|   | └─ pages/
|   |   Home.jsx
|   |   Upload.jsx
|   |   Dashboard.jsx
|   |   Validation.jsx
```

```
| | Anomaly.jsx
| | Quality.jsx
| | AIExplanation.jsx
| | Reports.jsx
| |
| | └─ services/
| |   api.js
| |   uploadApi.js
| |   reportApi.js
| |
| | └─ hooks/
| |
| | └─ utils/
| |
| | └─ context/
| |
| | └─ App.jsx
| |
| | └─ main.jsx
|
└─ package.json
└─ vite.config.js
```

2. Backend (Server)

Frontend pages

Landing Page



Upload Dataset



Dashboard



Validation Report



Anomaly Detection



Quality Score



AI Explanation

↓

Download Reports

3. Server (Node & Express)

server/

|

├─ config/

|

db.js

|

multer.js

|

├─ controllers/

|

uploadController.js

|

validationController.js

|

reportController.js

|

dashboardController.js

|

├─ routes/

|

uploadRoutes.js

|

validationRoutes.js

|

reportRoutes.js

|

dashboardRoutes.js

|

├─ middleware/

|

errorHandler.js

|

├─ models/

|

User.js

```
| Dataset.js
| Report.js
| Anomaly.js
|
|— services/
|   pythonService.js
|   reportService.js
|
|— uploads/
|
|— utils/
|
|— app.js
|
|— server.js
|
|— package.json
```

Backend APIs

POST

/api/upload

POST

/api/validate

POST

/api/anomaly

POST

/api/report

GET

/api/dashboard

GET

/api/report/:id

3.ML

ml_engine/

|

|— validation/

|

validator.py

|

missing.py

|

duplicate.py

- | datatype.py
- | date_validation.py
- | quantity_validation.py
- | price_validation.py
- | payment_validation.py
- | product_validation.py
- | status_validation.py
- | regex_validation.py
- | report.py
- | utils.py

- |

- | — cleaning/

- | cleaner.py
- | missing_handler.py
- | date_cleaner.py
- | payment_cleaner.py
- | product_cleaner.py
- | price_cleaner.py

- |

- | — rules/

- | business_rules.py
- | rule_engine.py

- |

- | — statistics/

- | zscore.py
- | iqr.py
- | outlier_engine.py

- |


```
└─ anomaly/
  │   isolation_forest.py
  │   local_outlier_factor.py
  │   dbscan_detector.py
  │   anomaly_engine.py
  │
  └─ scoring/
    │   dataset_score.py
    │   record_score.py
    │   completeness.py
    │
    └─ ai/
      │   gpt_summary.py
      │
      └─ reports/
        │   pdf_report.py
        │   excel_report.py
        │   ppt_report.py
        │
        └─ data/
          │
          └─ models/
            │
            └─ run_pipeline.py
            │
            └─ requirements.txt
```

ML Flow

CSV



Validation



Cleaning



Rule Engine



Statistical Detection



ML Detection



Quality Score



AI Summary



Generate Reports



Return JSON

Database

users

datasets

reports

anomalies

quality_scores

logs